

Integration of Facility Location and Hypercube Queuing Models in Emergency Medical Systems

Maryam Ghobadi,^a Jamal Arkat,^a Hiwa Farughi,^a Reza Tavakkoli-Moghaddam^b

^aDepartment of Industrial Engineering, University of Kurdistan, Sanandaj, Iran
m.ghobadi@eng.uok.ac.ir, j.arkat@uok.ac.ir (✉), h.farughi@uok.ac.ir

^bSchool of Industrial Engineering, College of Engineering, University of Tehran, Tehran, Iran
tavakoli@ut.ac.ir

Abstract. The purpose of emergency medical systems (EMS) is to save lives and reduce injuries with a quick response in emergencies. The performance of these systems is highly dependent on the locations of ambulances and the policy for dispatching them to the customers (i.e., patients). In this study, two new mathematical models are presented to combine the decisions about the location and dispatching policy by integrating the location and hypercube queuing models. In the presented models, the flow-balance equations of the hypercube queuing model are considered as the constraints of the location model. In the first model, the status of each server is idle or busy at any moment, as in the classic hypercube queuing model. In the second model, the travel time is considered independent of the on-scene time, and the status of each server is idle, busy, and traveling, or busy and serving a customer on the incident site. To verify the models, some small-sized examples are first solved exactly. Then, an optimization framework based on the genetic algorithm is presented due to the complexity of the models for solving larger-sized examples. Two approaches (i.e., the exact and simulation-optimization) are used to extend the framework. The results demonstrate that the proposed optimization framework can obtain proper solutions compared to those of the exact method. Finally, several performance measures are considered that can only be calculated using the second model.

Keywords: Emergency medical system, hypercube queuing system, simulation-optimization, genetic algorithm

1. Introduction

Emergency service systems (ESSs) are public service systems designed to save lives and reduce injuries from accidents. In these systems, customers are usually in critical unstable conditions. Therefore, any delay in serving them leads to death or serious injuries. Thus, these systems must provide services in the shortest time and with the highest quality. ESS design decisions can be classified into two categories: strategic and operational (Galvao and Morabito 2008). In strategic decisions, the number and locations of servers are determined. In operational decisions, however, policies are defined for how servers should be sent (dispatching strategies), how regions should be covered by servers, how large queue capacity should

be, and so on. These two categories depend on each other, and it can result in more efficient models to consider them simultaneously.

In recent years, one of the most significant criteria of social development has been the accessibility of medical facilities (Landa-Torres et al. 2013). Thus, the locations of ambulance stations and dispatching strategies have been particularly important. One of the most popular ESSs is the emergency medical system (EMS), in which a limited number of ambulances should be deployed at emergency stations to cover the demand areas in the best possible way. Furthermore, the issue has become more complicated due to high medical costs, increasing demand, and urban traffic congestion. These factors affect ambulances' service

times and should be considered in choosing the right places for them. Lee et al. (2020) simulated the healthcare system in Hong Kong to investigate the impact of such factors on optimal resource utilization.

ESSs can be divided into the customer-to-server and server-to-customer categories. In the first category, servers are fixed, and customers refer to them for service. In the second type, however, servers are sent to customers' locations. EMSs are of the server-to-customer type. When servers are fixed, they are usually not distinguishable, which means that it is not important which servers are busy; only the number of busy servers is critical (e.g., beds in a hospital or seats on a plane). Nevertheless, when servers move toward customers, it is possible to distinguish them from each other and define various features (e.g., priorities and average service time for each server), and even the workload of each server can be dependent on its location. The hypercube queuing model (HQM) introduced by Larson (1974) is a descriptive model for server-to-customer systems. The basic idea in this model is a descriptive expansion of the state space of a queuing system. Since most emergency systems are centralized, the HQM is very suitable for EMS analysis. In a centralized system, the customer contacts a central unit, which sends the first idle server according to the dispatching policies of the system and the priority list. A priority list can be created based on different criteria such as distance, server type, and customer demand type. Moreover, if an appropriate server is not available, the customer may join the queue or transfer to another emergency system.

The structure of this paper is as follows. Section 2 discusses some related literature. The problem is discussed in Section 3. In this section, two integrated mathematical models are presented. The performance of the mathematical models is evaluated through small ex-

amples in Section 4. An optimization framework based on GA in combination with an exact method and the simulation approach is presented in Section 5. Section 6 shows the computational results for several medium- and large-sized examples. Finally, Section 7 presents conclusions and suggestions for the future development of the model.

2. Related Literature

After Larson (1974), many researchers have used the HQM in their studies. Here is some of the most recent work done in the area. Rodrigues et al. (2017) prioritized customers in the queue in an HQM, where each server can only support certain other servers (partial backup). They also proposed an approximate algorithm to solve problems on a larger scale. Rodrigues et al. (2018) added the assumption of servers with different rates to this approximate algorithm and set different rates for customers who join and do not join the queue. Karimi et al. (2018) developed an approximation algorithm based on Larson's (Larson 1975) algorithm for an emergency system with partial backup and customer priorities. They calculated the service rate in their algorithm as a weighted average of the dispatch rates and travel times. Liu et al. (2020) defined server disruption based on allocating resources to each server. Then, to reduce the costs under server disruptions, they embedded the HQM into a mixed-integer nonlinear programming (MINLP) model to optimize resource allocation. Beojone et al. (2020) presented an HQM in which homogeneous servers located in a base are considered a group to reduce equilibrium equations. Blank (2020) extended a hypercube queuing approach for a large-scale study area and presented an algorithm to deal with the computational time by aggregating demand areas. Blank (2021) developed an HQM to consider virus outbreaks at refugee camps and evaluate the performance of emer-

gency systems during the pandemic. Liu et al. (2021) proposed a cooperative HQM with multiple dispatching. In their work, the state of emergency systems is described as a situation, in which vehicles in the same fleet simultaneously start and end services. Golabian et al. (2021) proposed an HQM to obtain the best return strategy for ambulances. In each state of their proposed model, the location of busy ambulances is not known and approximated based on the arrival pattern of customers. A comprehensive review of the literature was provided in Ghobadi et al. (2019) The focus of this research is only on papers that used the HQM along with location problems.

Daskin (1983) presented a maximum expected coverage location problem (MEXCLP) for the first time, in which servers are deployed to maximize expected coverage. He also considered the important basic assumption in his problem that servers may not be available. Batta et al. (1989) relaxed three basic assumptions in the MEXCLP and presented AMEXCLP. These assumptions include the independence of servers, equal busy probability for servers, and independence between the busy probability of servers and their locations. Galvao et al. (2005) compared the MEXCLP with the maximum availability location problem (MALP). MALP, first presented by ReVelle and Hogan (1989), maximizes the number of customers that can be covered within the standard time interval with reliability. They also developed extended MALP (EMALP), in which the busy probability of each server can be different. McLay (2009) considered several types of customers based on their demand prioritization (priorities 1, 2, and 3) in the MEXCLP and the proposed MEXCLP2. The objective was to maximize the number of customers in a life-threatening situation covered within the standard time interval. They provided an approximated algorithm based on the Jarvis algorithm Jarvis (1985) to calculate the busy

probability of servers in the case of the infinite queuing capacity. Davoudpour et al. (2014) presented a probabilistic coverage model for integrating the MEXCLP and the HQM. They examined the performance of the model at an EMS in Tehran. Barneveld et al. (2017) extended the MEXCLP2 model McLay (2009), and proposed a model to compute the compliance table for an EMS with two types of server. The authors used the hypercube model and iterative procedure proposed by McLay (2009) to estimate the input parameters of the model.

Geroliminis et al. (2004) combined a location model and a queuing system to minimize the mean response time and meet the minimum expected coverage level. They used the HQM to set the districting and dispatching policies. In their research, the server's response time was dependent on its location, and different service rates were considered for the customers located in the service area of that server and those outside that area. Later, Geroliminis et al. (2009) developed the above model further for locating emergency vehicles in an urban area. Geroliminis et al. (2011) presented an optimization algorithm based on the genetic algorithm (GA) to solve some examples of this model. Chanta et al. (2011) formulated the minimum p-envy location problem (MpELP) to locate ambulances for all customers to be served fairly. As a measure of fairness, envy is a function of the distance from a customer to its closest facility and the distance from a customer to its second-closest facility.

Toro-Diaz et al. (2013) presented a nonlinear mixed-integer optimization model to find the optimal locations of ambulances. In this model, the probabilities of steady states were calculated using the HQM. The results of their work indicated that the policy of dispatching the closest server to each customer minimizes response time and maximizes expected coverage. Then, Toro-Diaz et al. (2014) developed this model further and presented a

multi-objective location model to balance efficiency and justice. In their work, justice means equalizing the average response times and workloads of all servers. They used the algorithm provided by [Budge et al. \(2009\)](#), which is, in fact, an extension of the Jarvis algorithm, where more than one server can be assigned to each station. [Ansari et al. \(2015\)](#) used this approximation algorithm to estimate the correction factor, average server workload, and the workload of each server in an MILP model. [Ansari et al. \(2017\)](#) considered simultaneous dispatching of more than one server to serve a customer (i.e., multiple dispatching) when the severity of an accident is so high, or there is a need for special forces in addition to ordinary emergency assistance personnel at the accident site. [Yoon and Albert \(2018\)](#) presented an approximate hypercube model with a cut-off priority queue where lower-level customers are served only when the number of idle ambulances reaches the threshold number (cut-off level). They used this model within the MILP model of [Ansari et al. \(2015\)](#) to determine the locations of ambulances and dispatching strategies to respond to prioritized customers.

[Kim and Lee \(2016\)](#) presented two optimization algorithms based on HQM and simulation to find the locations of ambulance stations such that the expected reliability was met. They claimed that the results of the two algorithms were almost the same. [Rodrigues et al. \(2020\)](#) proposed a Mixed Integer Linear Programming (MILP) model for fire stations to maximize coverage. They used Jarvis' approximation algorithm to calculate server workload. Table 1 presents a list of studies in the literature with a focus on their assumptions [Ghobadi et al. \(2019\)](#).

As shown in Table 1, the HQM and location model are integrated into most existing studies. However, the only researchers who used the exact HQM in their location models are [Geroliminis et al. \(2004\)](#), [Geroliminis et al.](#)

[\(2009\)](#), [Geroliminis et al. \(2011\)](#), [Toro-Diaz et al. \(2013\)](#), and [Kim and Lee \(2016\)](#). A closer look at these studies revealed no connection in their works between the decision variables of the location models and steady-state probabilities. Flow-balancing equations are not explicitly considered in these models, and they can be considered at best as two-phase models, in which the locations of ambulances are determined in the first phase, and the states are analyzed in the next phase by the ambulance locations. In this study, two new mathematical models are presented, in which the flow-balance equations of the hypercube queuing model are considered as the constraints of the location models. In these models, the location and hypercube queuing models are integrated to combine the decisions about the location and dispatching policy. Therefore, this study first seeks to specify whether the integration of location and HQM can be effective in decisions on the locations of ambulances and improvement of performance measures.

Furthermore, this research seeks to investigate whether independent consideration of travel time and on-scene time has a significant impact on system performance measures and, as a result, on the optimal locations of ambulances. Given the importance of travel time as a major part of service time, several studies have also attempted to evaluate it more accurately. [Halpern \(1977\)](#) showed that there would be errors in calculating performance criteria even in a small system with two servers and two customers if a fixed service rate is considered. [Geroliminis et al. \(2011\)](#) estimated service rate by obtaining the harmonic mean of the service rates of inbound and outbound requests. [Iannoni et al. \(2011\)](#) assumed that on-scene service time is much longer than travel time and therefore ignored the effect of travel time on the performance measures of the system. [Karimi et al. \(2018\)](#) considered service rate as the weighted average of dispatch rates

Table 1 List of Studies in the Literature with a Focus on Their Assumptions

Reference	Hypercube		Location model	Objective function	Solution approach	HQM as an optimization	Case study
	Exact	Approximate					
Batta et al. (1989)		*	AMEXCLP	Maximum expected	Daskin's heuristic		
Geroliminis et al. (2004)	*			Minimum mean response time	An iterative heuristic		*
Galvao et al. (2005)		*	EMALP	Maximum number covered of customers with reliability α	Simulated annealing and pure vertex substitution (VS) local search heuristic		
McLay (2009)		*	MEXCLP2	Maximum number of high-level customers covered within a threshold	Branch and bound algorithm		*
Geroliminis et al. (2009)	*			Minimum mean response time	Discrete-event simulation		*
McLay and Mayorga (2010)		*		Maximum number of customers covered in a time threshold	-		*
Geroliminis et al. (2011)	*			Minimum mean response time	An embedding heuristic of GA with the hypercube model	*	*
Chanta et al. (2011)		*		Minimum total weighted	Tabu search algorithm		*
Toro-Diaz et al. (2013)	*			Minimum mean response time	Genetic algorithm		*
Davoudpour et al. (2014)	*		MEXCLP	Maximum emergency region coverage	Branch and bound algorithm		*
Toro-Diaze et al. (2014)		*		Minimum mean response time and maximum fairness	Reactive tabu search (RTS)		*
Ansari et al. (2015)		*		Maximum coverage	An iterative heuristic		*
Kim and Lee (2016)	*		PLSCP	Minimum number of ambulances	Two iterative algorithms based on hypercube and simulation	*	
Van Barneveld et al. (2017)		*		Maximum covered demand within a time threshold	Exact algorithm and simulation		*
Ansari et al. (2017)		*		Maximum expected coverage	An iterative heuristic and simulation		*
Yoon and Albert (2018)		*		Maximum expected coverage	An iterative heuristic and simulation		*
Rodriguez et al. (2020)		*	MILP	Maximum covered demand	An iterative heuristic based on the Jarvis algorithm	*	*

and travel times. However, it is practically impossible to easily calculate the average of on-scene and travel times for real-world instances. The on-scene time depends on factors (e.g., the severity of the incident, the degree of expertise of the emergency team, and the associated equipment); however, travel times depend on factors (e.g., the type of vehicle, the route, and the severity of traffic). Furthermore, travel time takes a significant portion of service time, and ignoring it will lead to wrong decisions.

3. Problem Presentation and Mathematical Models

As mentioned earlier, strategic and tactical decisions have to be made for designing an ESS.

The strategic decisions determine the number and locations of ambulances, and the tactical decisions determine the ambulance dispatching policy and the areas covered by each ambulance. These two levels are interdependent, and the decisions made at each level strongly affect the other. The main purpose of this study is to provide models for EMSs that make the strategic and operational decisions more integrated. For this purpose, the facility location and HQM are combined. According to the existing constraints, location models seek the best locations for the deployment of emergency stations and allocate ambulances to them. On the other hand, the HQM analyzes the states of the system, which are themselves dependent

on the locations of ambulances. Based on this, the first model is presented, in which the location variable and par flow-balance equations are integrated. The difference between this model and the existing models in the literature is that the decision variables play an important role here in the flow-balance equations of the queuing system. Therefore, flow-balance equations are explicitly considered as model constraints, and the dispatching policy and areas covered by each ambulance are effective in determining the locations of ambulances. Another purpose of this study is to consider the travel time individually. The service time starts from the moment a server is assigned to a customer, and it lasts until the server returns to its base. Thus, service time includes the travel time from the server base to the customer's location, time of service at the customer's location (on-scene time), and travel time from the customer's location to the server base. Usually, service time follows the exponential distribution, and in most studies in the area of emergency, only one rate is used to indicate average service time. In this research, both travel time and on-scene time follow exponential distributions; however, their rates are independent and different. A new state definition is presented, in which the status of a server at any moment is one of the following statuses: idle and staying at its station, traveling from the station to the customer's location or vice versa, and providing service at the site of the incident. Then, the flow-balance equations of the queuing system are formed based on the new state definition and integrated into a location model.

The presented models are mixed-integer nonlinear optimization ones. In the first model, as in the classic hypercube, two statuses are considered for each server (i.e., zero and one), which show that a server is free or busy, respectively. In the second model, travel time is considered individually; therefore, three statuses are considered for each server: idle, trav-

eling, and serving on-scene.

3.1 Assumptions

The entire area is divided into several smaller regions called atoms, and the center of each atom is considered as the location of customers. The arrival process of demands is generated via the Poisson process with a different rate for each atom. Furthermore, potential points for deployment of emergency stations are located at the center of each atom, and an ambulance is assigned to a station once located. On arrival of a customer, the closest available ambulance is dispatched (single dispatching) if there is one within the predetermined standard distance (partial backup). This means that only ambulances deployed at a standard distance to a customer (coverage radius) can serve him. The queue capacity is assumed to be zero. Thus, when a customer arrives while all the ambulances within his coverage radius are busy, that customer is lost or referred to another EMS. Also, when an ambulance completes its service to a customer, it first returns to its base and is then dispatched to serve another customer. In Sections 4.1 and 4.2, two mathematical models are provided according to these assumptions.

The proposed models concern an EMS meeting the following assumptions:

1. Servers are mobile and distributed spatially.
2. The area is divided into several atoms, and emergency stations can be located at the center of each atom.
3. One ambulance is assigned to each station if it is deployed.
4. When a customer arrives, the closest available server within the coverage radius is dispatched to that customer.
5. The priority list of servers is fixed for each customer and is defined based on the distances between the servers and the customer.

The arrival process of customers is the Poisson process with a different rate for each atom.

The notations are listed below.

Sets and indices:

N Total number of servers, $n = 1, \dots, N$.

J Set of demand points, $j = 1, \dots, J$.

I Set of candidate nodes for deploying facilities, $i = 1, \dots, I$.

R Set of system states, $r = 1, 2, 3, \dots$

Parameters:

λ_j Arrival rate of demand point j .

λ Total arrival rate.

μ Service rate in Model I.

δ On-scene service rate in Model II.

t_{ij} Distance between demand point j and the center of atom i .

γ_{ij} The travel rate between demand point j and the center of atom i .

h_i The status of the server deployed at atom i .

Sets of system states: M_r Set of servers that are idle in state r .

W_r Set of servers that are busy in state r .

K_i Set of atoms that can be covered by the server located at atom i .

S_{ij} Set of candidate nodes that are closer to demand point j than candidate node i .

Decision variables: ρ_j Probability that a customer from demand point j is served by a server.

$P(r)$ Probability of being in state r .

x_i 1, if a facility is deployed at candidate node i ; 0 otherwise.

y_{ijr} 1, if server i is the closest idle server to demand point j in state r , 0 otherwise.

3.2 Model I with Two Statuses for Each Server

It is necessary to calculate the steady-state probability of the system to evaluate the performance measures of an EMS, such as the average response time, the average workload, the number of calls serviced by the highest-priority server, and the number of lost customers. For

the specification of the steady-state probabilities, the system of flow-balance equations must be solved. These equations are determined by the flow-balance criterion around each state of the system. In this model, service time follows the exponential distribution with rate μ for each server, as in classic HQM. Therefore, each state of the system is represented by a list of free and busy servers (an array of 0s and 1s). For example, state $\{100\}$ corresponds to a 3-server system, in which server 1 is busy, and servers 2 and 3 are free (reading from left to right). Based on the flow-balance criterion, the rate at which the system enters the state i in a steady state is equal to the rate at which the system leaves that state. As server status changes from idle to busy or vice versa, the system leaves its current state and enters another. A transition occurs when each state is entered or left. In the transition rate flow diagram, each transition occurs over an edge of a hypercube.

For proposing an integrated model, it is necessary to describe how the flow-balance equations are written. Equation 1 shows these equations for the queuing system:

$$\begin{aligned}
 P(r) & \left\{ \sum_{i \in M_r} \sum_{j \in K_i} \lambda_j + \sum_{i \in W_r} \mu \right\} \\
 & = \sum_{i \in M_r} \mu P \left(R: h_i = 0 \rightarrow 1 \right) \\
 & \quad + \sum_{i \in W_r} \sum_{j \in K_i} \lambda_j P \left(R: h_i = 1 \rightarrow 0 \right) \quad (1)
 \end{aligned}$$

The left-hand side of the equation shows the probability of exiting state r of the system. In the first case, in which one of the idle servers is busy (upward transition), the exiting rate is represented by $\sum_{i \in M_r} \sum_{j \in K_i} \lambda_j$. The rate indicates that for a server to be dispatched to a customer, firstly, the server must be idle ($i \in M_r$), and secondly, the customer must be within its coverage radius ($j \in K_i$). In the second case,

one of the busy servers ($i \in W_r$) may complete the service and go back to its base (downward transition). The right-hand side of the equation represents the probability of entering the state r . In this equation, $(R:1 \rightarrow 0)$ indicates all the states with downward Hamming distances equal to 1 from the state r ; similarly, $(R:1 \rightarrow 0)$ shows all the states with upward Hamming distances equal to 1 from the state r . The Hamming distance between two vertices is the number of digits that are different between them. Upward Hamming distance concerns the number of idle servers (0) in the current state, the statuses of which are changed to busy (1) in adjacent states. Downward Hamming distance pertains to the number of busy servers (1) in the current state of the system, the statuses of which are changed to idle (0) in other states.

Equation (1) represents a system of flow-balance equations. It should be noted in the case of this system, of course, that the sum of all the probabilities must be equal to one (Equation (4)). For these equations to apply to a location model, there must be a relationship between the state probabilities and location decision variables. For this purpose, the term that indicates the service completion of a server in each equation must also contain the decision variable associated with the deployment of that server (facility). Thus, the term should be multiplied by the decision variable x_i so that the relevant clause will be ineffective if the server is not deployed ($x_i = 0$). On the other hand, if a transition occurs as a customer arrives, the server must be deployed, idle, and have the highest priority for that customer. To this end, in addition to Constraint (2), y_{ijr} must be multiplied by the arrival rate of each customer in the flow-balance equation, as suggested by its definition. Constraint (2) ensures that the closest deployed facility covers each customer. Therefore, the following

three constraints must be added to the location model. In this equation, if facility i is the closest to customer j , where the server is idle in state r , then x_i and, consequently, y_{ijr} are set to 1. However, if there is at least one established facility in S_{ij} , which is closer to the customer arriving from atom j than the facility located at atom i , then $\sum_{d \in S_{ij}, d \in M_r} x_d$ is greater than or equal to 1, and the existence of this constraint becomes ineffective.

$$y_{ijr} \geq x_i - \sum_{d \in S_{ij}, d \in M_r} x_d, \forall i \in M_r, j \quad (2)$$

$$\begin{aligned} & P(r) \left\{ \sum_{i \in M_r} \sum_{j \in K_i} (\lambda_j y_{ijr}) + \sum_{i \in W_r} (\mu x_i) \right\} \\ &= \sum_{i \in M_r} \mu P(R:0 \rightarrow 1) \times x_i \\ &+ \sum_{i \in W_r} \sum_{j \in K_i} P(R:1 \rightarrow 0) (\lambda_j y_{ijr}), \forall r \quad (3) \end{aligned}$$

$$\sum_{r \in R} P(r) = 1 \quad (4)$$

The formulation of Model I is presented below.

$$\min Z = \sum_{j=1}^J \frac{\lambda_j}{\lambda} (1 - \rho_j) \quad (5)$$

Subject to:

$$\sum_{i=1}^I x_i = N \quad (6)$$

$$\sum_{i=1}^I y_{ijr} \leq 1, \forall j, r \quad (7)$$

$$y_{ijr} \geq x_i - \sum_{d \in S_{ij}, d \in M_r} x_d, \forall i \in M_r, j, r \quad (8)$$

$$\begin{aligned}
& P(r) \left\{ \sum_{i \in M_r} \sum_{j \in K_i} (\lambda_j y_{ijr}) + \sum_{i \in W_r} (\mu x_i) \right\} \\
&= \sum_{i \in M_r} \mu P(R:0 \rightarrow 1) \times x_i \\
&+ \sum_{i \in W_r} \sum_{j \in K_i} P(R:1 \rightarrow 0) (\lambda_j y_{ijr}), \forall r \quad (9)
\end{aligned}$$

$$\sum_{r \in R} P(r) = 1 \quad (10)$$

$$\rho_j = \sum_{r \in R} P(r) \sum_{i=1}^I y_{ijr}, \forall j \quad (11)$$

$$y_{ij}, x_i \in \{0, 1\}, P(r) \in (0, 1) \quad (12)$$

Objective function (5) minimizes the probability of lost arriving demands. Equation (6) ensures that the number of servers assigned to a facility does not exceed the maximum number of available servers. Equation (7) denotes that each customer is assigned to only one deployed server in each state of the system. Constraint (8) ensures that the closest deployed facility covers each customer. Equation (9) shows the flow-balance equations, and Equation (10) ensures that the sum of probabilities is equal to one. In Equation (11), ρ_j is calculated. Constraint (12) refers to the domain of decision variables.

3.3 Model II with Three Statuses for Each Server

As mentioned earlier, travel time is considered independent of on-scene time in the second model (Model II); they both follow the exponential distribution; however, their distributions are independent and have different rates. The rate of on-scene time is fixed for all servers, and the rate of travel time depends on the distance between the server and the customer. In this HQM, each server can be in one of the following three statuses at any moment: idle and stationed at its base (0), busy and traveling

from the base to the incident site, or vice versa (h'_i), and busy and serving on-scene (h_i). h_i is the number of the demand point to which the server has been sent to serve a customer. For example, state $\{012'\}$ corresponds to a 3-server system, in which server 1 is idle, server 2 is busy on the scene and serving a customer having arrived from atom 1, and server 3 is busy and traveling to or from atom 2 (reading from left to right). As in the previous section, the method of forming the flow-balance equations is described below. Equation (13) shows these equations for this HQM.

In the first expression on the left-hand side, the summation of the arrival rates of customers who can receive service from this system in state r is shown. A customer can receive service only if there is at least one idle server within its coverage radius ($i \in M_r$ and $j \in K_i$). In the second expression on this side, the total rate of on-scene time is shown for a server serving a customer at his location in state r ($i \in W_r$). The last term on this side calculates the total rate of travel time for the servers that are traveling ($i \in L_r$). The travel status can indicate that the server is moving from its base to the customer's location, and it can also indicate that the server is returning from the customer's location to its base; therefore, the expression is multiplied by $1/2$.

The probability of entering state r from other states is calculated on the right-hand side of Equation (13). In the first term on this side, the total travel rate is calculated for the servers, which are idle in state r ($i \in M_r$) but have been returning to their bases from a customer's location in the previous state. Of course, the Hamming distance between these two states must be one, and all servers must have the same status in them except for one that is idle in r but has been traveling in the previous state. The coefficient of $1/2$ is also considered for reasons similar to those mentioned earlier. The total travel rate for servers, which have been serv-

ing on the scene before but are returning to their bases in state r , is calculated in the second term on this side. The Hamming distance must be one also in this case, and the statuses of all servers are the same except for one that is traveling in state r but has been serving at a customer's location. The last two parts on this side calculate the total service rate and total arrival rate from states with a Hamming distance of one to state r , respectively. The servers traveling to their bases in state r have been serving on the scene in the past, so they enter state r at rate δ . Furthermore, the servers that are traveling to the customer's location in state r have been idle in the past and are sent to the customer's location upon his arrival, so this transition occurs at rate $\lambda_{|h_i|}$, in which $|h_i|$ shows the number of demand points that the customer has entered.

$$\begin{aligned}
 P(r) & \left\{ \sum_{j \in \bigcup_{i \in M_r} K_i} \lambda_j + \sum_{i \in W_r} \delta + \sum_{i \in L_r} \frac{1}{2} (\gamma_{i,|h_i|} + \gamma_{|h_i|,i}) \right\} \\
 &= \frac{1}{2} \sum_{i \in M_r} \sum_{k \in K_i} \gamma_{i,k} P(R: h_i = 0 \rightarrow k') \\
 &+ \frac{1}{2} \sum_{i \in W_r} \gamma_{i,|h_i|} P(R: h'_i \rightarrow h_i) + \sum_{i \in L_r} \delta P(R: h'_i \rightarrow h_i) \\
 &+ \sum_{i \in L_r} \lambda_{|h_i|} P(R: h'_i \rightarrow 0) \quad (13)
 \end{aligned}$$

For Equation (13) to apply to Model II, the decision variables must be included in this equation, as in the previous section. Therefore, Equation (14) is written as follows, and to formulate a location model, one would need only to use Equation (14) instead of Equation (9) in Model I. There are $\prod_{i=1}^I (2|k_i| + 1)$ flow-balance equations, in which $|k_i|$ is the number of atoms that can be covered by the server located at atom i , if the base of this atom is deployed. Some other performance criteria, like the ratio of time when servers are traveling to or from demand points, out of their atom (TR), the proportion of time when servers are busy serving customers (TS), and the proportion of

time when servers are busy serving their closest customers (TC), and many other criteria presented in Ghobadi et al. (2020), can be calculated only in Model II. These three measures are calculated using Equations (15)-(17).

$$\begin{aligned}
 P(r) & \left\{ \sum_{j \in \bigcup_{i \in M_r} K_i} (\lambda_j y_{ijr}) + \sum_{i \in W_r} (\delta x_i) + \sum_{i \in L_r} \frac{1}{2} (\gamma_{i,|h_i|} + \gamma_{|h_i|,i}) x_i \right\} \\
 &= \frac{1}{2} \sum_{i \in M_r} \sum_{k \in K_i} \gamma_{i,k} x_i P(R: h_i = 0 \rightarrow k') \\
 &+ \frac{1}{2} \sum_{i \in W_r} \gamma_{i,|h_i|} x_i P(R: h'_i \rightarrow h_i) + \sum_{i \in L_r} \delta x_i P(R: h'_i \rightarrow h_i) \\
 &+ \sum_{i \in L_r} \lambda_{|h_i|} y_{ijr} P(R: h'_i \rightarrow 0) \quad (14)
 \end{aligned}$$

$$TR = \sum_i \frac{\sum_{r: i \in L_r \& |h_i| \neq i} P(r)}{\sum_{r: |h_i| \neq 0} P(r)} * x_i \quad (15)$$

$$TS = \sum_i \frac{\sum_{r: i \in W_r} P(r)}{\sum_{r: |h_i| \neq 0} P(r)} * x_i \quad (16)$$

$$TC = \sum_i \frac{\sum_{r: i \in W_r, |h_i| = i} P(r)}{\sum_{r: |h_i| \neq 0} P(r)} * x_i \quad (17)$$

4. Case Study

Model 1 can be considered as an extension of the model presented by Toro-Diaz et al. (2013). In their research, they tried to evaluate the performance criteria of the emergency system in two stages according to the selected locations for the establishment of facilities. In Model 1, the evaluation of the system performance criteria is performed in an integrated manner. To test the performance of Model 1 compared to the model presented by Toro-Diaz et al. (2013), the random instances presented in their work referred to as Example 1 from here on, are first formulated using the model I under three different scenarios. Then, these instances are solved exactly because their solution spaces are small enough to be enumerated completely. It is assumed here that there

are five atoms, and the center of each atom is the location of demand points. Moreover, these centers are considered candidate sites for the deployment of emergency stations. Table 2 shows the total demand arriving from each atom and the coordinates of the center of each atom. The center of each atom is also a potential location for servers. As the model I is integrated, Baron, a GAMS solver, is used for solving these examples. The Baron solver is a computational system for non-convex optimization problems, which guarantees optimal solutions. The goal is to choose three candidate sites from among five and establish a station at each selected site to minimize the probability of losing customers. The coverage radius is 7 km, and the right angle distances between atom centers are obtained according to the coordinates, as shown in Table 3. In this table, the distance between the potential location considered in each atom for the deployment of a base and the demand point of the atom is 0.1 km.

Table 2 Center Coordinates and Demands of Atoms

Atom	x-coord	Y-coord	demand
1	10	5	20
2	1	1	18
3	7	9	12
4	2	7	8
5	6	1	6

Table 3 Travel Distance between Atoms (KM)

From \ to	1	2	3	4	5
1	0.1	9.84	5	8.24	5.65
2	9.84	0.1	10	6.08	5
3	5	10	0.1	5.38	8.06
4	8.24	6.08	5.38	0.1	7.21
5	5.65	5	8.06	7.21	0.1

As in Toro-Diaz et al. (2013), these examples are solved for different values of $\rho = \frac{\lambda}{3\mu}$, for sensitivity analysis and validation of Model I. For a constant demand rate and $\rho = 0.1, 0.5$, and 0.9 , the service rate is 213.3, 42.6, and 23.7, respectively. The results are shown in Table 4. In this table, x represents the stations established

in the optimal state, and z is the probability of losing customers given the stations' locations. Saturation refers to the probability that all the servers are busy, $P[1, 1, 1]$. Moreover, St. Coverage in this table refers to the classic definition of coverage. If it is 1, at least one ambulance base is established at a distance of 7 or less from each demand point. As can be seen in the table, the model is well suited to the problem. When the service rate of the servers has decreased, the probability of losing customers has increased.

For further comparison between the integrated model presented in this research and that presented in Toro-Diaz et al. (2013), the probability of losing customers according to their suggested locations is also shown in this table as z' . As can be seen, the integrated model I functions more properly than their two-stage model. Of course, the improvement observed in the solution of a problem of this size is expected to be due in part to the more accurate method of calculating the performance criteria in the integrated model. As mentioned earlier, all the examples in this section are small-sized and can therefore be solved accurately within relatively short times (about 0.1 Sec.).

Table 4 Optimal Solution Information

μ	x	Z	Saturation	St. Coverage	z'
213.3	1,4,5	0.021	0.002	1	0.4
42.6	1,4,5	0.204	0.086	1	0.234
23.7	1,4,5	0.370	0.214	1	0.398

For examination of the accuracy of model II and its sensitivity to the input data, Example 1 is formulated using this model. To formulate the problem with Model II, it is necessary to determine the travel rate between different atoms. The distances between the centers of atoms are shown in Table 3. The speed of each server is 1 km/h, as assumed in Toro-Diaz et al. (2013), so the travel time between two atoms is calculated by dividing the distance between them by speed. Accordingly, the rate of travel time,

which is equal to the inverse of travel time, is calculated as in Table 5. The results of this example, formulated with model II, are shown in Table 6. The columns in this table are the same as those in Table 4, but the calculation of saturation probability is slightly different. This probability is equal to the sum of the probability values of states in which all the servers are busy in either on-scene or traveling status. As is clear from Table 6, this problem is impractical due to the low travel rates, the servers are always in travel, and almost all the customers are lost. Therefore, the effect of considering travel time can be seen easily, even in this example. It is clear from a comparison of the contents of this table to the results in Table 4 that considering travel time is effective on decisions on the optimal locations for emergency bases. Since this is a strategic decision, it is essential to take travel time into account. As is clear from Table 6, in addition to the difference between the two models in optimal locations for facility deployment, the probability of losing customers has increased, which was expected since the servers are considered to be busy at the times of traveling.

Table 5 Rates of Travel Time between Atoms in Example 1

From \ to	1	2	3	4	5
1	10	0.101	0.2	0.121	0.176
2	0.101	10	0.1	0.164	0.2
3	0.2	0.1	10	0.185	0.124
4	0.121	0.164	0.185	10	0.138
5	0.176	0.2	0.124	0.138	10

Table 6 Optimal Solution Information

μ	x	Z	Saturation	St. Coverage
213.3	1,2,5	0.9946	0.9862	1
42.6	1,2,5	0.9946	0.9863	1
23.7	1,2,5	0.9947	0.9863	1

For further evaluation of the performance of the models, the speed of the servers is considered to be 50 km/h. There are also different values for demand rate and on-scene service time. The results are shown in Table 7 and

Table 8. As is clear from Table 7, the problem is well formulated by model 2, and the results are perceptible. According to these results, the probability of losing customers will be decreased if the service rate increases or the arrival rate decreases. It is also clear from a comparison of the contents of this table to the results in Table 2 that considering travel time is effective in decision-making on the optimal locations for emergency bases, and it is important to take travel time into account since this is a strategic decision. The performance measure values of this system under different demand rates and on-scene service rates are shown in Table 8. Unfortunately, the solution time of this model is extremely high, and the larger the problem size, the longer the running time. This is due to the increasing number of flow-balance equations, making it difficult or even impossible to achieve the exact solution. In the following, other ways to solve this kind of problem are introduced.

Through a comparison of the results of the two presented models in this table and the previous tables, it can be concluded that model II generally exhibits greater stability and less sensitivity to the inputs.

5. Optimization Framework Based on GA

As mentioned earlier, the presented models are nonlinear and contain state variables as well as location and allocation variables. In a model with I candidate sites, 2^I state variables and, consequently, 2^I constraints (flow-balance equations) are added to Model I, and $\prod_{i=1}^I (2|k_i| + 1)$ states and equations are added to Model II. As the number of candidate sites increases, the numbers of variables and constraints are also increased greatly. Thus, there is a need for methods that can solve medium and large examples of the models in a reasonable time. Many studies have emphasized the effectiveness of metaheuristic methods (e.g.,

Table 7 Results of Solving Example 1 Formulated with Model II and an Increased Travel Rate

μ	λ							
	{20, 18, 12, 8, 6}		{12, 12, 12, 12, 12}		{2, 1.8, 1.2, 0.8, 0.6}		{20, 6, 6, 6, 6}	
	x	z	x	z	x	z	x	z
213.3	1,2,3	0.761	2,3,4	0.812	1,2,3	0.129	1,2,4	0.678
42.6	1,2,3	0.798	1,3,4	0.834	1,2,3	0.169	1,2,4	0.721
23.7	1,2,3	0.822	1,3,4	0.851	1,2,3	0.205	1,2,4	0.752

Table 8 Performance Measures in Model II

μ	λ											
	{20, 18, 12, 8, 6}			{12, 12, 12, 12, 12}			{2, 1.8, 1.2, 0.8, 0.6}			{20, 6, 6, 6, 6}		
	TR	TS	TC	TR	TS	TC	TR	TS	TC	TR	TS	TC
213.3	2.8	0.13	0.07	2.86	0.1	0.03	2.56	0.25	0.2	2.81	0.12	0.07
42.6	2.4	0.54	0.3	2.57	0.39	0.17	2.01	0.86	0.66	2.43	0.51	0.29
23.7	2.11	0.84	0.46	2.33	0.51	0.21	1.7	1.2	0.89	2.15	0.8	0.44

GA) for obtaining good solutions in an appropriate time.

The GA was introduced by Holland (1975) and became popularized later by Goldberg (1989). It is a general population-based algorithm that is usually successful in the achievement of the optimal solution. Many studies have confirmed that it is a powerful algorithm in solving combinatorial problems (Geroliminis et al. 2011, Iannoni et al. 2008).

Aytug and Saydam (2002) introduced the genetic algorithm as a successful algorithm for solving combinatorial problems. Goldberg (2004) also pointed out in his research that a combination of the GA with spatial queuing systems (the hypercube model) or simulation will achieve very good results in dealing with real-world issues concerning emergency systems. Jia et al. (2007) demonstrated the efficiency of the genetic algorithm in solving the problem of locating facilities in large-scale emergency systems. Toro-Diaz et al. (2013) stated that the genetic algorithm has always been capable of achieving the same or pretty close solutions to those obtained by the enumeration procedure in a joint location-HQM model. The GA is easy to understand and flexible. Also, this algorithm enables the exploitation of previous or alternate solutions. How-

ever, to take full advantage of the GA, it is necessary to choose its parameters according to the characteristics of the problem. According to Aytug and Saydam (2002), there are five main items to be selected in designing the GA: 1) the appropriate solution representation structure (i.e., Chromosome), 2) an effective mutation operator, 3) a proper crossover operator, 4) an initial feasible solution, and 5) appropriate rates for crossover and mutation and adequate population size.

5.1 Solution Representation and Tuning the GA

The solution representation in a location model is a simple zero-one chromosome; however, it is troublesome and leads to unfeasible solutions to use this type of chromosome straightly. Thus, random numbers between 0 and 1 are generated. Each chromosome consists of I genes (i.e., a number of candidate sites), and a randomly generated number is assigned to each gene. For conversion of these chromosomes to zero-one arrays, the N larger genes are selected and assigned number 1, and the other genes are assigned number 0. Number 1 indicates that a station is deployed at the candidate site. Figure 1 shows a representation of a chromosome. According to this chromosome, sites 1, 2, and 5 are selected, and stations are

established at them.

Through this procedure, some chromosomes are generated randomly to make up the initial population. Therefore, the initial population contains feasible solutions. To create a new generation from the current population, the following three steps should be considered.

Selection: $(1 - f_1)Q$ chromosomes are selected from the current solution, and added to the new population, where Q is the number of chromosomes in each population, and f_1 represents the percentage of the population that should be replaced in each iteration using the crossover operator. The probability of choosing each chromosome (CR) is as follows.

$$P(CR_g) = \left(\sum_g \text{Fitness}(CR_g) \right) / \text{Fitness}(CR_g) \quad (18)$$

Single-point crossover: Crossover is performed on the parent chromosomes, which contain random numbers. For this purpose, a random integer between 1 and I is generated, which is then used as the separation point of the selected parents.



Figure 1 Representation of a Chromosome

Mutation: for this operation, f_2 percent of the chromosomes from the new population are selected, and the mutation operator is performed on them. For maintaining the feasibility of the chromosomes, a swap is done between two genes, one of which has a numeric value of one while the other is zero. A generation of ineffective solutions is thus prevented.

GA parameter selection. For setting the parameters of the GA, several examples are solved with different values for the crossover (P_c) and mutation (P_m) rates. The number of iterations for each instance is determined such that the product of the number of iterations

and population size remains constant (approximately 10,000). For example, if the population size is 30, the number of iterations will be 334. The results indicate that $P_m = 1/L$ (L represents the length of the chromosome) and $P_c = 0.5$ are proper values. The results also demonstrate that for small examples, $[0.75I]$, which returns the largest integer smaller than $0.75I$, gives an appropriate population size. For medium and large-sized examples, a population size of 100 is also suitable. Finally, the numeric value of the fitness function obtained from the GA is compared to that obtained from the exact method. The gap between these two solutions is calculated by Equation (19).

$$\text{Gap} = \frac{\text{GaSolution} - \text{BestSolution}}{\text{BestSolution}} \quad (19)$$

A positive value for the gap indicates that the solution obtained from the GA is worse than the best possible solution provided by the exact method. If the gap is zero, both methods have obtained the same solution, and a negative value for the gap is probably due to the stochastic nature of the parameters and approximations in the calculations.

5.2 Chromosome Evaluation (Encoding)

As mentioned in the previous section, the way of evaluation of the chromosomes is very important in the GA. In this study, two methods are suggested to evaluate chromosomes, the exact approach, and the simulation-optimization approach.

5.2.1 Exact Approach

The first suggestion is to accurately solve flow-balance equations by the Gaussian elimination method (GEM) to evaluate each chromosome. In the GEM, the system of equations is reduced by elimination operations to another system with the same solutions. The reduction process includes three operations: row and column displacement, replacement of each row with a multiple of itself, and summation of coefficients of two rows, and replacement of one

of the two. Each step aims to select one of the matrix elements and turn all the elements below that into zero using the three operations. The coefficient matrix is gradually transformed into an upper triangular matrix. The advantage of this method is that although the entire solution space is not considered, each chromosome is examined precisely. On the other hand, the disadvantage of this method is that the generation of the system of equations is time-consuming and requires a large amount of memory for larger-sized examples.

5.2.2 Simulation-Optimization Approach

The second suggestion is to use the simulation-optimization approach. This approach seeks to optimize an objective function for its constraints, and the objective function or constraints, or both, can be evaluated through simulation. A simulation is a common approach to evaluating real large-scale systems with complexity (Chen and He 2005), and unlike mathematical programming, which is model-driven, this approach does not require a model and only uses input data to evaluate objective functions and constraints. Most simulation-optimization approaches are dependent only on input and output data for finding the optimal solution. Unlike the Gaussian elimination method, this approach solves problems of any size, but its random nature affects the final output, and the gap between this method and the exact one is sometimes not zero, even for small-sized examples. Huang et al. (2018) used a simulation-optimization model based on dynamic analysis to evaluate the performance of a railway system and showed that there is a relationship between travel time and energy-consuming.

For the use of simulation-optimization methods, it is necessary to provide a more precise definition of the problem to achieve more accurate solutions. Therefore, four basic points should be considered: 1) determining the input variables, 2) scaling and reducing the range

of decision variables, 3) providing proper initial assumptions for the algorithm, and 4) using benchmarks for better estimation (Amaran et al. 2016). Simulation-optimization methods can be divided into different categories according to solution space and variable type (Amaran et al. 2016). Because of the stochastic nature of the problem on hand, the simulation used in this study is a discrete event simulation. Furthermore, it can be referred to as optimization with simulation iterations since it combines simulation and optimization. The GA is also a well-known algorithm in the area of simulation, which is used in much simulation-optimization software (Amaran et al. 2016), and many studies have shown that a combination of simulation and GA can improve service quality in an EMS.

Table 9 and Table 10 illustrate the simulation procedure used in this study for Models 1 and 2, respectively. The procedures are used for the evaluation of the chromosomes, which are generated by the GA. Thus, whenever a chromosome has to be evaluated, the simulation function is called to simulate the queuing system according to the stations deployed on that chromosome. Because of the stochastic nature of simulation, however, the fitness function of the chromosome takes a different value each time it is called. Although the difference between these outputs is not great, the simulation process is run 50 times for improvement of the decisions based on the function, and the average output is considered as the cost function value of the chromosome. In Table 9, $EL1$ is a state vector with N (i.e., the number of servers) elements, which are 0s and 1s. If the k^{th} element in the vector is zero, the k^{th} server is idle; otherwise, it is busy. Moreover, t_A and t_S indicate the next arrival time of a customer and the next service completion time of a server, respectively. W_{EL1} is a set of servers that are busy based on $EL1$. In Table 10, t_A is the customer's arrival time, t_C is the on-

scene service time, t_{gf} shows the traveling time of the server to reach the customer's location, t_{cb} is the traveling time of the server to return from the customer's location to the station, and $EL2$ is an N -dimensional vector that indicates the current statuses of servers. Therefore, each $EL2$ member can take the following values: 0 (idle), r_i (serving at the customer's location), r_i'' (traveling to the customer's location), and r_i''' (returning from the customer's location), where r_i indicates the atom number to which the server has been sent to serve. W_{EL2} is the set of servers that are serving a customer based on $EL2$. L_1 is the set of servers that are traveling to the customer's location from their stations, and L_2 indicates the set of servers that are traveling from the customer's location to their stations given $EL2$. In Table 9 and Table 10, T represents the total simulation run time, and the value of 100,000 hours is selected. For larger values of T , only the run time of simulation has increased, and the values of the performance measures have remained virtually unchanged. At all the stages, U is a new random number.

6. Computational Results

In this section, first, example 1 is solved again for efficiency validation of the simulation-optimization approach. The gap between the results of solving these examples using the exact method (Models I and II) and those of the simulation-optimization approach are shown in Table 11. As shown in this table, the gap which cannot be solved exactly. Following, examples with larger dimensions are designed to further analyze Model I and II using optimization approaches. between the exact methods and simulation-optimization approaches is less than 1.6%. Therefore, it can be concluded that the proposed approaches are efficient and can be used for solving large examples.

6.1 Real-World Case Study: Hanover County

To investigate the capability of model I, real-world data from Hanover Fire and EMS department is formulated with this model. The emergency system in this county covers 474 square miles and 122 demand zones. The total demand rate is, on average, 1.2 calls/h. To find optimal locations that minimize the possibility of losing customers, the hybrid GA-GEM, and simulation-optimization approach. The results are shown in Table 12. The average gap for solving this example with two proposed approaches based on GA is about 0.5%, which shows the good performance of these two methods in achieving the solution.

Table 12 Probability of Losing a Customer in the Case Study

μ	GEM		Simulation	
	N=6	N=7	N=6	N=7
1.2	0.1438	0.1298	0.1568	0.13
0.86	0.1852	0.1583	0.1909	0.1601

6.2 Medium and Large-Sized Case Study



Figure 2 Network of the Candidate Locations for Designing Examples

For designing these examples, the network provided in Figure 2 is used. The distance between every two points is defined as the nearest road distance, as shown in Table 13. The coverage radius is 7 km, and the on-scene service rate is 10. In this example, the speed of the vehicle is 50 km/h. In Table 14, x is the best location for deployment of a station as proposed by the GA, z^s is the probability of losing customers, which is obtained through the

Table 9 Simulation Procedure for Model I

Initialization
$t_A = \frac{-1}{\sum_j \lambda_j} \ln U ; t_S = T; \quad EL1 = \{0, 0, \dots, 0\}.$
First case: if $t_S < t_A$ and $t_S \leq T$, $t = t_S$. Specify the first server, the service of which is finished. Change the state of the selected server from zero to one in the state vector. $t_S = t - \frac{1}{\sum_{k \in W_{EL1}} \mu_k} \ln U .$
Second case: otherwise, if $t_A < T$, $t = t_A$. Specify the node of the arriving customer. If there is at least one idle server in a standard distance to that customer, do: Specify the closest idle server; Change the state of the selected server from zero to one in the state vector; $t_S = t - \frac{1}{\sum_{k \in W_{EL1}} \mu_k} \ln U ;$ $t_A = t - \frac{1}{\sum_j \lambda_j} \ln U .$ Otherwise, increment the number of lost customers by one.
Third case: otherwise, Obtain the ratio of the number of lost customers to the total number of simulated customers.

Table 10 Simulation Procedure for Model II

Initialization
$t_A = \frac{-1}{\sum_j \lambda_j} \ln U ; t_C = T; t_{cb} = T; t_{gf} = T \quad EL2 = \{0, 0, \dots, 0\}.$
First case: if $t_C < t_A$, $t_C \leq t_{gf}$, $t_C \leq t_{cb}$ and $t_C \leq T$, $t = t_C$ Specify the server that will finish its task sooner than the set of servers that are busy serving customers (W_{EL2}). In the state vector, change the status of this server from r_i to r''_i . $t_C = t - \frac{1}{\sum_{k \in W_{EL2}} \mu_k} \ln U$ $t_{cb} = t - \frac{1}{\sum_{k \in L_2} \gamma_{ r_i k}} \ln U$
Second case: if $t_{gf} < t_A$, $t_{gf} \leq t_C$, $t_{gf} \leq t_{cb}$ and $t_{gf} \leq T$, $t = t_{gf}$ Specify the server that will finish its task sooner than the set of servers that are busy traveling to customers' locations from their stations (L_1). In the state vector, change the status of this server from r''_i to r_i . $t_{gf} = t - \frac{1}{\sum_{k \in L_1} \gamma_{k, r_i }} \ln U$ $t_C = t - \frac{1}{\sum_{k \in W_{EL2}} \mu_k} \ln U$
Third case: if $t_{cb} < t_A$, $t_{cb} \leq t_C$, $t_{cb} \leq t_{gf}$ and $t_{cb} \leq T$, $t = t_{cb}$ Specify the server that will finish its task sooner than the set of servers that are busy and traveling to their station from the customer's location (L_2). In the state vector, change the status of this server from r'''_i to 0. $t_{cb} = t - \frac{1}{\sum_{k \in L_2} \gamma_{ r_i k}} \ln U$ $t_C = t - \frac{1}{\sum_{k \in W_{EL2}} \mu_k} \ln U$
Fourth case: if $t_A < t_{gf}$, $t_A < t_{cb}$, $t_A \leq t_C$ and $t_A \leq T$, $t = t_A$ Specify the atom where the customer arrives. Increment the total number of customers by one. Identify the closest idle server to that customer. If there is such a server, change its status from 0 to r''_i . $t_A = t - \frac{1}{\sum_j \lambda_j} \ln U$ $t_{gf} = t - \frac{1}{\sum_{k \in L_1} \gamma_{k, r_i }} \ln U$ If there is no such a server, increment the total number of lost customers by one.
Fifth case: Otherwise, Obtain the ratio of the number of lost customers to the total number of simulated customers.

Table 11 Percentages of the Gap Between the Exact Method and the Simulation-Optimization

Model	Number of Examples															Average
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
I	0	0.34	0.21	0.35	0.46	0.22	0.81	0.1	0.2	0.1	0.21	0.2	0.71	0	0	0.26
II	0.51	0	0.1	-0.37	0.15	0.05	0.77	-0.21	0.65	0.52	0	-0.8	-0.5	-0.99	-0.6	0.13

GA in combination with the Gaussian Elimination method, and z^s shows this probability as obtained through the proposed simulation-optimization method. Furthermore, in this table, ρ^g and ρ^s indicate the probability that all servers are busy as calculated by the above two methods.

Figure 3 shows the impacts of increasing the on-scene service rate and decreasing the travel time rate on TR and TS for problems of different sizes. As can be seen in this figure, service time decreases as the rate of on-scene time increases, so the servers are relatively longer in traveling status. Similarly, travel time increases as travel rates decrease, and servers are in on-scene status for relatively shorter periods.

7. Conclusion

One of the most important issues in designing an emergency medical system (EMS) is determining optimal locations for ambulance stations. Specification of appropriate locations for stations is effective on the system performance measures, such as service time and expected coverage, and more lives are saved as a result.

In this research, a location model was proposed for emergency systems that minimized the number of lost customers. Here, a customer would be lost if there was no idle server on his arrival within the standard distance to him. The states of the system are described by a hypercube queuing model (HQM). There had to be a relationship between the state probabilities and location variables. For this purpose, an integrated model was proposed, in which the flow-balance equations of the HQM were regarded as the constraints of the location model, and the location variables were part of

the flow-balance equations. With this model, the locations of ambulance stations were determined along with the assignment of ambulances to customers.

The extended model was nonlinear, in which the numbers of variables and constraints had grown significantly. Therefore, an optimization framework needed to be developed. The small-sized examples of the model were solved with exact methods; however, an optimization framework based on the GA was proposed for solving the larger-sized examples. Although the GA does not ensure that the global optimal solution is obtained, many studies have confirmed the capability of this algorithm to find proper, near-optimal solutions. In this study, two approaches were proposed for the evaluation of the chromosomes in the GA. The first approach was to use the Gaussian elimination method to solve the flow-balance equations exactly and calculate state probability accurately. Despite its high accuracy, this method could find the optimal solutions only to small and medium-sized examples (up to 10000 flow-balance equations) within a reasonable time, and run time was too high for larger examples. The second approach used the simulation-optimization method. Since it did not consider the flow-balance equations, this approach solves large-sized examples within a short time. The accuracy of the method was examined with a comparison of the results to those of the exact method. The average gap between the two methods was under 2%, with little variance.

Although other methods, such as the approximate hypercube, are also available, there is no approximate algorithm that is consistent with all the proposed model's assumptions.

Table 13 Distance Matrix between the Candidate Locations in Figure 2

	1	2	3	4	5	6	7	8	9	10	11
1	1	5	13	12	14	12	18	28	27	37	36
2	5	1	8	9	11	7	13	23	22	32	31
3	13	8	1	11	13	15	15	25	36	26	38
4	12	9	11	1	2	6	4	14	13	29	21
5	14	11	13	2	1	4	3	13	11	21	20
6	12	7	15	6	4	1	7	17	15	25	24
7	18	13	15	4	3	7	1	10	14	25	17
8	28	23	25	14	13	17	10	1	10	15	7
9	27	22	36	13	11	15	14	10	1	10	20
10	37	32	26	29	21	25	25	15	10	1	12
11	36	31	38	21	20	24	17	7	20	12	1

Table 14 Results for the Proposed GA-Gaussian Elimination, and GA-Simulation-Optimization

Number of servers	Rate of travel time	Demand rate	x	z^g	ρ^g	z^s	ρ^s
$N = 4$	$\gamma_{ij} = \frac{50}{t_{ij}}$	$\lambda = \{11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1\}$	1,2,3,5	0.759	0.296	0.769	0.300
		$\lambda = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$	5,9,10,11	0.751	0.161	0.758	0.169
		$\lambda = \{5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5\}$	2,4,5,7	0.749	0.291	0.758	0.130
	$\gamma_{ij} = \frac{0.2 \times 50}{t_{ij}}$	$\lambda = \{11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1\}$	3,5,8,9	0.887	0.177	0.889	0.177
		$\lambda = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$	8,9,10,11	0.851	0.296	0.852	0.301
		$\lambda = \{5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5\}$	8,9,10,11	0.855	0.130	0.855	0.130
$N = 7$	$\gamma_{ij} = \frac{50}{t_{ij}}$	$\lambda = \{11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1\}$	1,2,3,4,5,6,7	-	-	0.618	0.087
		$\lambda = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$	5,6,7,8,9,10,11	-	-	0.611	0.081
		$\lambda = \{5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5\}$	5,6,7,8,9,10,11	-	-	0.601	0.078
	$\gamma_{ij} = \frac{0.2 \times 50}{t_{ij}}$	$\lambda = \{11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1\}$	3,5,7,8,9,10,11	-	-	0.842	0.132
		$\lambda = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$	3,5,7,8,9,10,11	-	-	0.803	0.130
		$\lambda = \{5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5\}$	3,5,7,8,9,10,11	-	-	0.783	0.120

Therefore, it is suggested as a basis for future research that an approximate algorithm be provided that considers travel times independently in addition to the partial backup. Considering uncertain parameters, such as demand rates that can depend on the hour of the day or day of the week, can make the problem closer to the real world. In these conditions, the arrival process is generated via the non-homogenous Poisson process, and the system of equations cannot be formulated; however, simulation-optimization methods can be used. Moreover, the definition of policies for dividing demand areas into sub-areas and examining the effect of these policies on performance measures (e.g., expected coverage and response time) are proper suggestions for further development of the model. In most studies in the literature, every time the service of a

server is completed, it should return to its base and be dispatched to a new customer even if there is a customer in the queue, while the elimination of the returning time can help to reduce response time. Besides, in all studies conducted so far, service time has been the sum of travel time and on-scene service time, and a single rate is considered for total service time. However, travel time depends on factors (e.g., traffic and type of vehicle), and on-scene service time depends on factors (e.g., the severity of the patient's conditions and the capability of the servers). Thus, it does not seem accurate to consider a single unit rate for total service time, and it makes the problem closer to the real world to take into account different rates for travel time and on-scene service time. It will also help improve the system's performance measures to dispatch multiple servers simulta-

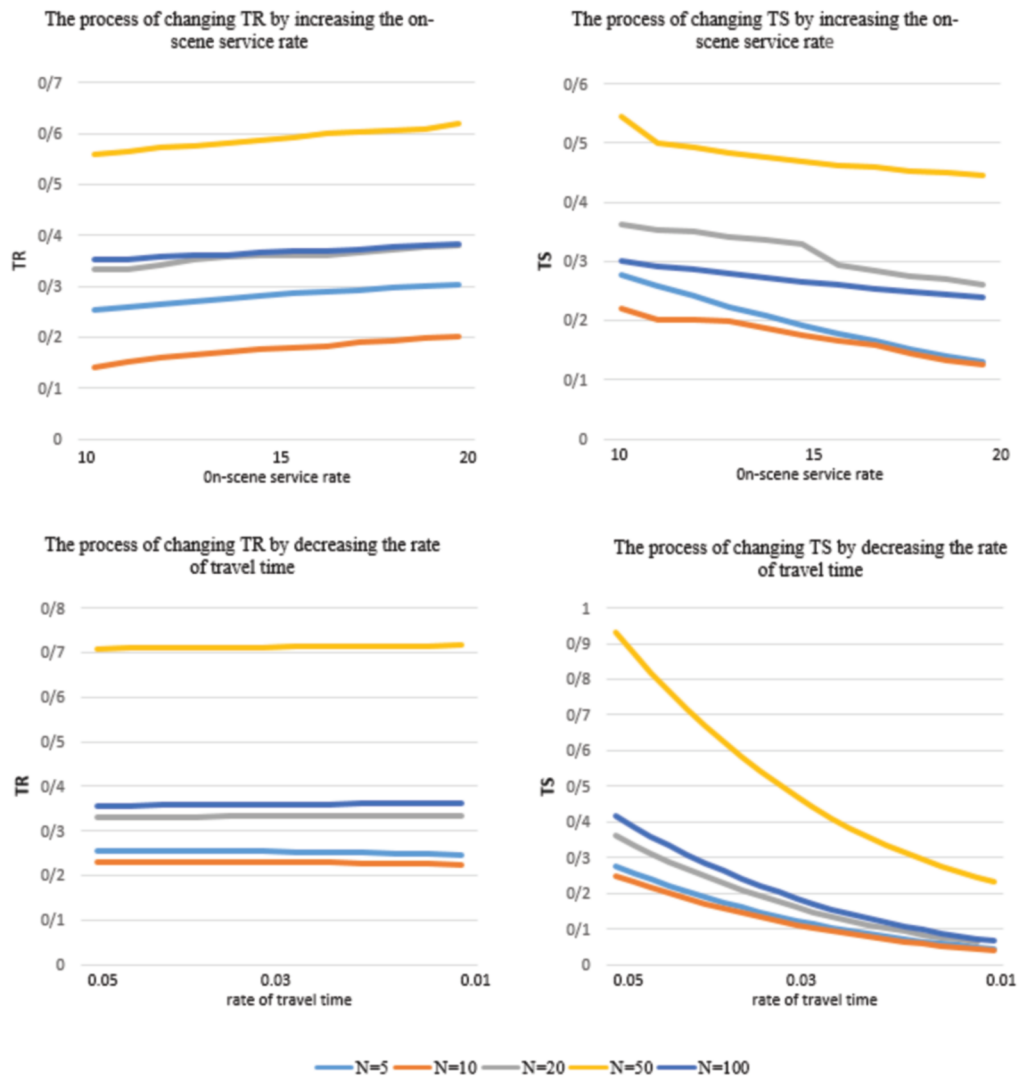


Figure 3 Process of Changing TR and TS by Increasing On-Scene Service Time (the First Row) and Decreasing the Rate of Travel Time (the Second Row)

neously according to the customer's demands or to consider stations that contain multiple servers.

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- Maryam Ghobadi** is the Head of Industrial Engineering Department at the Shomal University, Amol, Iran. She obtained her Ph.D., M.Sc. and B.Sc. degrees in industrial engineering from the Kurdistan University in Sanandaj (2021), Alzahra University in Tehran (2013), and the Iran University of Science and Technology in Behshahr (2009), respectively. She has been teaching in this field from 2013 at the undergraduate and post graduate level. She has published a book, a book chapter, and more than 10 journal and conference papers.
- Jamal Arkat** received B.Sc., M.Sc., and Ph.D. degrees in industrial engineering from Iran University of Science and Technology (IUST) in Iran. He is currently an associate professor at the Department of Industrial Engineering at the University of Kurdistan, Sanandaj, Iran. His research interests include operations research applications in health care, crisis management, and stochastic processes.
- Hiwa Farughi** is an associate professor at the University of Kurdistan. He was born in 1976. He received B.Sc. and M.Sc. degrees in industrial engineering from Amir Kabir University of Technology in 1998 and 2000, respectively. He received his Ph.D. degree in industrial engineering from Iran University of Science and Technology in 2012. His research interest topics include production planning, decision making techniques and quality control.
- Reza Tavakkoli-Moghaddam** is a professor of industrial engineering at the College of Engineering, University of Tehran, Iran. He obtained his Ph.D., M.Sc. and B.Sc. degrees in industrial engineering from the Swinburne University of Technology in Melbourne (1998), the University of Melbourne in Melbourne (1994), and the Iran University of Science and Technology in Tehran (1989), respectively. He serves as the Editor-in-Chief of "Advances in Industrial Engineering" journal published by the University of Tehran and as the editorial board member of nine reputable academic journals. He is the recipient of the 2009 and 2011 Distinguished Researcher Awards and the 2010 and 2014 Distinguished Applied Research Awards at the University of Tehran, Iran. He has been selected as the National Iranian Distinguished Researcher in 2008 and 2010 by the MSRT (Ministry of Science, Research, and Technology) in Iran. He has obtained an outstanding rank as the top 1% scientist and researcher in the world elite group since 2014. He also received the Order of Academic Palms Award as a distinguished educator and scholar for the insignia of Chevalier dans l'Ordre des Palmes Academiques by the Ministry of National Education of France in 2019. He has published 5 books, 32 book chapters, and more than 1000 journal and conference papers.